Sardar Patel Institute of Technology

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(Autonomous College of Affiliated to University of Mumbai)

Re-Examination
August 2023

Maxi Marks: 100
Class: T.E
Course code: CS301/IT301
Name of the course: Theory of Computation

Duration: 3 hours
Semester: V
Branch: COMP/IT

Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Q No		Max Marks	CO	BL
Q1	Attempt all the questions			
a	Which two of the following four regular expressions are equivalent? Justify your answer. (i) (00)* (ε + 0) (ii) (00*) (iii) 0* (iv) 0(00)* (i) and (ii) B) (ii) and (iii) C) (i) and (iii) D) (iii) and (iv)	2	2	3
b	How many states the smallest finite automation will have one that accepts the language L = {x | length of x is divisible by 3}. Justify your answer.	2	1	3
c	Let ω be any string of length n in {0, 1}*. Let L be the set of all substrings of ω. What is the minimum number of states in a non-deterministic finite automaton that accepts L? (A) n -1 (B) n (C) n +1 (D) 2^{n+1}	1	1	3
d	Design FSM which contains NO occurrence of aa over ∑ (a, b). **OR** Construct the NFA to recognize the string ending with abb.	5	1	3
Q2 a	State and prove Arden's theorem with example.	10	2	3
Q2 b	Design Mealy m/c for 2's complement of a given binary number.	10	1	3
Q3 a	Construct a DFA with reduced states equivalent to the regular expression: 10+(0+11)0*1	10	1	3
Q3 b	Give and explain formal definition of Pumping Lemma for Regular Language and prove that the following language is not regular. L = {WW | W ϵ {a, b}*}.	10	2	3
Q4 a	Design a Turing Machine to recognise all strings consisting of even number of 1's. Assume the string is made up of only 1's, i.e. the input is a unary number.	10	4	3

Q4 b	Construct a PDA for the language $\{a^n b^{2n} \mid n>=1\}$ OR Prove that NPDA ≠ DPDA	10	4	3
Q5 a	Convert the following CFG into CNF S→ aSa | bSb |a |b |aa |bb	10	3	3
Q5 b	Convert the following CFG into GNF S→AB A→BS| b B→SA| a	10	3	3
Q6 a	Construct a grammar for L= $\{a^j b^n c^n \mid n>=1, j>=0\}$	05	3	3
Q6 b	Describe Type 2 and Type 3 grammar with example.	05	3	3